

# Pressure in fluids and in the atmosphere

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

|        | QUESTION                                                                          | ANSWER                                                                                                         | ✓                        |
|--------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------|
| 1      | Name the two states of matter that are considered fluids.                         | Liquid and gas                                                                                                 | <input type="checkbox"/> |
| 2      | Describe what causes pressure in a fluid.                                         | The particles of the fluid colliding with surfaces.                                                            | <input type="checkbox"/> |
| 3<br>★ | A fluid exerts a force on its container. State the direction of the force.        | At right angles to the surface.                                                                                | <input type="checkbox"/> |
| 4      | State the equation linking pressure, area and force.                              | pressure = force ÷ area<br>( $p = F \div A$ )                                                                  | <input type="checkbox"/> |
| 5<br>★ | State the standard unit of pressure.                                              | Pascal (Pa)                                                                                                    | <input type="checkbox"/> |
| 6<br>★ | Describe how the pressure in a liquid depends on the height of the liquid column. | The taller the column, the greater the pressure.                                                               | <input type="checkbox"/> |
| 7      | Describe how the pressure in a liquid depends on its density.                     | The denser the liquid, the greater the pressure.                                                               | <input type="checkbox"/> |
| 8<br>★ | Explain what causes upthrust.                                                     | Pressure increases with depth, so the pressure on the bottom of a submerged object is greater than on the top. | <input type="checkbox"/> |
| 9      | Describe the Earth's atmosphere.                                                  | A thin layer of air around the Earth.                                                                          | <input type="checkbox"/> |
| 10     | Describe how the density of the atmosphere changes with height.                   | It decreases as height increases.                                                                              | <input type="checkbox"/> |
| 11     | Give the cause of atmospheric pressure.                                           | Air molecules colliding with a surface.                                                                        | <input type="checkbox"/> |
| 12     | Describe how atmospheric pressure changes with height.                            | It decreases as height increases.                                                                              | <input type="checkbox"/> |

| QUESTION |                                                                 | ANSWER                                                                           | ✓                        |
|----------|-----------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------|
| 13       | State the equation for the pressure due to a column of liquid.  | pressure = height × density × gravitational field strength<br>( $p = h \rho g$ ) | <input type="checkbox"/> |
| 14<br>★  | Explain why the pressure in a liquid increases with depth.      | The deeper you go, the greater the weight of liquid above.                       | <input type="checkbox"/> |
| 15       | State the condition for an object to float.                     | Upthrust equals its weight.                                                      | <input type="checkbox"/> |
| 16       | State when an object in a liquid sinks.                         | When its weight is greater than the upthrust.                                    | <input type="checkbox"/> |
| 17<br>★  | Explain why atmospheric pressure decreases as height increases. | There is less air above, so fewer air molecules collide with each unit of area.  | <input type="checkbox"/> |